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Tapwave Best Practices

Current Version	Date	Author
REV 0.3	07/27/2003	MS
REV 0.4	07/28/03	RR

Revision History

Version	Date	Description	Author
REV 0.1	7/14/2003		RR
REV 0.2	7/15/2003		RR
REV 0.3	7/27/2003	LIGHT COPY EDIT & FORMATTING CHANGES	MS
REV 0.4	7/28/2003	ADD PLACEHOLDER FOR UPLOAD REQUIREMENTS	RR

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The purpose of this document is to describe the development practices recommended by Tapwave. By following these recommendations, your application will provide your users with a common experience and will make effective and efficient use of the unique Tapwave features.

1. Sound

- Respect the Game Sound Level system preference. Use `TwSndSetVolume` and `TwSndGetVolume` if you provide your own user interface for managing the overall game volume. See the *Tapwave Advanced Sound API Reference* document for more information.
- Add support for Mute and Base Boost (when appropriate). If you provide your own user interface for managing the overall game volume, use `TwSndSetMute` and `TwSndSetBassBoost` respectively. See the *Tapwave Advanced Sound API Reference* document for more information.

2. High Score

- Only call the High Score Conduit when a new high score has been achieved. The High Score Conduit posts an application's **highest** score to Tapwave.com. For efficient use of network and battery resources, it is only necessary to call the High Score Conduit when a new, highest score has been achieved.
- Add support for tournament mode and decide whether your tournament is based on time, score, number of kills, etc. The user should access the tournament functionality through a *Tournament* menu item. Be sure to disable cheats and turn off any other game features that could give users an advantage over other users when tournament mode is enabled. See "High Score" in the *Tapwave Programmer's Reference* for more information.
- Register a description of your game and it's creator ID at the Tapwave Developer Zone (the URL will be provided at a later date). This is necessary to correctly map your game with its creator ID as reported by the high score conduit.

3. Controls

- A game should not override the behavior of the *Home* button. Pressing the Home button should always quit the current game and return the user to the launcher (also called the Home screen).
- A game should not override the behavior of the power button or the Bluetooth button.

- Game controls are split into two functional areas: Game Play and Game Menus.
- Control within the menus must be consistent across all games and follow these guidelines:

Action	Result
Tapping the screen or pressing the Function button	Brings up the game menu.
Manipulating the joystick or tapping the screen	Navigates through the game menu.
Pressing the Action "Up" button or tapping the screen	Selects the highlighted item in the game menu.

Control within the game should be consistent across all games when possible. Offer users the ability to either fully customize controls or select between preset control templates. Tapwave provides templates for several common game play types; including: Driving, Sports (Offense), Sports (Defense) and First Person Shooter. Not all games will map nicely to the templates listed below, therefore developers should feel free to modify these if a specific game requires something different.

Driving Games

Joystick	Steer
Blue (up) Action Button	Switch cameras
Yellow (right) Action Button	Reverse
Green (down) Action Button	Accelerate / Forward
Red (left) Action "Left" Button	Brake
Left Trigger	Select weapon (Offensive or Defensive)
Right Trigger	Fire weapon

Sports Games (Offensive)

Joystick	Move / Turn
Joystick Select	Jump / Dive
Blue (up) Action Button	Icon Pass
Yellow (right) Action Button	Pivot

Green (down) Action Button	Pass
Red (left) Action Button	Shoot / Kick / Swing
Left Trigger	Guard
Right Trigger	Turbo
Function Button	Call a play

Sports Games (Defensive)

Joystick	Move / Turn
Joystick Select	Jump / Dive
Blue (up) Action Button	Icon Switch
Yellow (right) Action Button	Steal / Intercept
Green (down) Action Button	Switch player
Red (left) Action Button	Block / Rebound / Catch

First Person Shooter

Joystick	Look
Joystick Select	Crouch
Blue (up) Action Button	Move forward
Yellow (right) Action Button	Move right
Green (down) Action Button	Move backwards
Red (left) Action Button	Move left
Left Trigger	Shoot
Right Trigger	Select
Function Button	Jump

4. Game Flow

- A game should support the following loading sequence when launched:
  - Display a splash screen then display a progress indicator if loading the game takes longer than 2 seconds.
  - Display the main game menu.
- If appropriate for your game, save the state of the game when quitting. This allows your users to return to where they left off. If your game does not support saving the game state, then notify users before quitting.
- A game should provide users with the option of quitting a game without saving the game state. (For instance, if they want to begin a new game.)

5. Interacting with the OS/Hardware

- A game must call `EvtGetEvent` (or `TwNextEvent`) on a regular basis to ensure that the system operates normally (e.g.: responds to alarms, low power, ExgManager events, etc). Your game must pause game play and work correctly under these conditions.
- A game should support rumble effects when appropriate. Don't overuse the rumbler as it puts additional drain on the battery.
- A game should support multi-player Bluetooth when possible. *(More guidelines to come.)*
- A game should support 2-player IR when possible.
- A game should support full-screen landscape mode (320x480). If appropriate, you may want to display the status bar during game play. (Displaying the status bar reduces the screen size to 320x450).
- A game should store game data as a separate file rather than within the game PRC -- this ensures the most efficient use of system memory. Use the following guidelines for managing your game's data files:

  All game data should be stored on a VSF volume (either on an external memory card or on the internal card) using the following path: `"/PALM/Programs/<prcdatabasename>-<creatorid>/"`, where `<prcdatabasename>` is the Data Manager's database name of the main application PRC file; (the *appl* file). For instance, if your application's database name is "Application" and the creator id is `"Strt"`", then the path would be `"/PALM/Programs/Application-Strt/"`. The following functions are defined to help facilitate path creation: `TwGetDBDataDirectory` and `TwGetPRCDataDirectory`. You can override the default path name by adding a `'Twdp'` resource with ID `#0` to your main application file. This resource should specify the name to use in place of `"<prcdatabasename>-<creatorid>"`. You should only override this if you need to share data between applications.

Following this convention reduces clutter and allows the Tapwave launcher to copy game data correctly between the internal memory and an external memory card. The game data follows the game PRC if the PRC is moved from internal memory to external memory and visa versa.

Note that each file on a VFS volume is a Palm OS database as are all the directories. You should never delete a VFS directory database, you could corrupt the card.

Game preferences should still be stored using the Palm OS Preferences API, the above information only applies to game data such as graphics and sound files.

6. Tapwave.com

<placeholder, information to come later>

7. References

PalmSource provides additional best practices documentation. Refer to the *Zen of Palm* at http://www.palmos.com/dev/support/docs/zenofpalm/ZenTOC.html and *Palm OS User Interface Guidelines* at http://www.palmos.com/dev/support/docs/ui/UIGuidelinesTOC.html.